

Module 3: Data Link Control & Error Handling Protocols

Topics 20 to 27 • Framing, CRC Polynomial Division, Hamming Codes & ARQ Protocols

Module III: Data Link Control & Error Handling Protocols — 8-Topic Master Syllabus Guide

Topic 20: Asynchronous vs. Synchronous Data Transmission (Start/Stop Bits vs Clock Framing)

Topic 22: Types of Errors (Single-Bit Errors vs. Burst Errors & Error Probability)

Topic 24: Cyclic Redundancy Check (CRC Polynomial Division, Modulo-2 Math & Generator Polynomials)

Topic 26: Hamming Code Encoding & Syndrome Decoding Algorithm ((7, 4) Hamming Code Trace)

Topic 21: Framing Protocols (Character Count, Byte Stuffing / Character-Oriented, Bit Stuffing HDLC)

Topic 23: Error Detection Techniques (Simple Parity, 2D Parity, Internet Checksum 1's Complement)

Topic 25: Error Correction & Linear Block Codes (Hamming Distance $d_{\min}$, Error Detection & Correction Limits)

Topic 27: Flow & Error Control Protocols (Stop-and-Wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ & Efficiency η)

Topic 20 & 21: Transmission Framing & Byte/Bit Stuffing

Framing Method	Operating Mechanism	Key Vulnerability & Solution
1. Character Count	A header field specifies the total number of characters in the frame.	Catastrophic Desynchronization: A single bit error in the count field prevents framing of all subsequent frames!
2. Byte Stuffing (Character-Oriented)	Frames begin and end with special 'FLAG' bytes ('0x7E'). Any appearance of 'FLAG' inside data is escaped by inserting an 'ESC' byte ('0x7D').	Overhead dependent on data content; tied to 8-bit character boundaries (PPP Protocol).
3. Bit Stuffing (Bit-Oriented HDLC)	Frames bounded by flag pattern '01111110' (six consecutive 1s). The sender automatically inserts (stuffs) a '0' after any sequence of five consecutive 1s in data.	Hardware Standard: Transparent bit transmission; receiver strips any '0' following five consecutive '1's (HDLC, SDLC).

Solved Problem: Bit Stuffing in HDLC Protocol

Original Payload Data: '011111101111100111111110'

Sender Stuffed Bitstream: (Stuffs '0' after every five '1's):

Payload: 0 11111010 11111000 11111011100

Enclosed Frame Transmitted: '01111110' [Stuffed Bits] '01111110'

Topic 23 & 24: Error Detection & Cyclic Redundancy Check (CRC)

CRC Modulo-2 Binary Division Algorithm: 1. Let Dataword have k bits D , and Generator polynomial have degree r ($r + 1$ bits G). 2. Append r zero bits to Dataword: $D \cdot 2^r$. 3. Perform Modulo-2 binary division (using **XOR** operations) of $D \cdot 2^r$ by G . 4. The resulting r -bit remainder is the **CRC Checksum (R)**. 5. The Codeword transmitted is: $T = D \cdot 2^r + R$. 6. **Receiver Verification:** Divide received frame T' by G . If remainder is **0**, accept frame; otherwise, error detected!

Step-by-Step Solved Problem: CRC Generation and Error Detection

Dataword $D = 1010000$ ($k = 7$ bits). Generator Polynomial $G(x) = x^4 + x + 1 \implies G = 10011$ (degree $r = 4$).

Step 1: Append $r = 4$ zeros: $D \cdot 2^4 = 10100000000$.

Step 2: Modulo-2 Division:

```
      1011001 (Quotient)
10011 | 10100000000
      10011
      ----
        11100
        10011
        ----
          11110
          10011
          ----
            11010
            10011
            ----
              10010
              10011
              ----
                00010 (Bring down remaining 0) → Remainder = 0110
```

Transmitted Codeword: $T = 10100000110$.

Topic 25 & 26: Error Correction & The (7, 4) Hamming Code

Hamming Distance & Correction Capability Theorems: - To **detect d bit errors**: Minimum Hamming Distance must satisfy: $d_{\min} \geq d + 1$. - To **correct t bit errors**: Minimum Hamming Distance must satisfy: $d_{\min} \geq 2t + 1$. - **Redundancy Bits Formula:** For m data bits and r parity bits: $2^r \geq m + r + 1$.

Step-by-Step (7, 4) Hamming Code Encoding and Syndrome Error Correction

Encode Dataword $D = 1011$ ($m = 4$ bits) into a 7-bit Hamming codeword ($r = 3$ parity bits at positions 1, 2, 4):

- Bit positions: $P_1(1), P_2(2), D_3(3), P_4(4), D_5(5), D_6(6), D_7(7)$
- Place data bits: Bit 3 = 1, Bit 5 = 0, Bit 6 = 1, Bit 7 = 1.
- $P_1 = D_3 \oplus D_5 \oplus D_7 = 1 \oplus 0 \oplus 1 = 0$
- $P_2 = D_3 \oplus D_6 \oplus D_7 = 1 \oplus 1 \oplus 1 = 1$
- $P_4 = D_5 \oplus D_6 \oplus D_7 = 0 \oplus 1 \oplus 1 = 0$
- **Transmitted Codeword: 0 1 1 0 0 1 1.**

Receiver Error Detection: Suppose bit 5 flips during transit $\implies$ Received: 0110111.

$$\text{Syndrome } S_1 = b_1 \oplus b_3 \oplus b_5 \oplus b_7 = 0 \oplus 1 \oplus 1 \oplus 1 = 1$$

$$\text{Syndrome } S_2 = b_2 \oplus b_3 \oplus b_6 \oplus b_7 = 1 \oplus 1 \oplus 1 \oplus 1 = 0$$

$$\text{Syndrome } S_4 = b_4 \oplus b_5 \oplus b_6 \oplus b_7 = 0 \oplus 1 \oplus 1 \oplus 1 = 1$$

$$\text{Error Position} = S_4 S_2 S_1 = (101)_2 = 5 \implies \text{Flip Bit 5 to correct error!}$$

Topic 27: Sliding Window Flow Control Protocols & Channel Efficiency

Protocol	Sender Window Size (W_S)	Receiver Window Size (W_R)	Sequence Numbers (k bits)	Channel Efficiency (η)
Stop-and-Wait ARQ	$W_S = 1$	$W_R = 1$	1 bit (0, 1)	$\eta = \frac{1}{1 + 2a}$
Go-Back-N (GBN) ARQ	$W_S = N = 2^k - 1$	$W_R = 1$ (Discards out-of-order)	k bits ($0 \dots 2^k - 1$)	$\eta = \frac{N}{1 + 2a}$ (for $N \leq 1 + 2a$)
Selective Repeat (SR)	$W_S = N = 2^{k-1}$	$W_R = N = 2^{k-1}$ (Buffers out-of-order)	k bits	$\eta = \frac{N}{1 + 2a}$ (for $N \leq 1 + 2a$)

Bandwidth-Delay Product & Parameter a Formula:

$$a = \frac{T_{\text{prop}}}{T_{\text{trans}}} = \frac{\text{Distance/Propagation Speed}}{\text{Frame Size/Bandwidth}}$$

Step-by-Step Solved Problem: Satellite Link Protocol Efficiency Comparison

A geostationary satellite channel has a one-way propagation delay $T_{\text{prop}} = 250$ ms, bandwidth $B = 1$ Mbps, and frame size $L = 1000$ bytes (8000 bits). Calculate channel efficiency for: (1) Stop-and-Wait, (2) Go-Back-N ($k = 4$ bits), and (3) Selective Repeat ($k = 4$ bits).

Solution:

$$T_{\text{trans}} = \frac{8000 \text{ bits}}{1,000,000 \text{ bps}} = 8 \text{ ms}$$

$$a = \frac{T_{\text{prop}}}{T_{\text{trans}}} = \frac{250 \text{ ms}}{8 \text{ ms}} = 31.25 \implies 1 + 2a = 1 + 2(31.25) = \mathbf{63.5}$$

- **1. Stop-and-Wait:** $\eta = \frac{1}{63.5} = 0.0157 = \mathbf{1.57\%}$ (Extremely poor channel utilization!).
- **2. Go-Back-N ($k = 4 \implies W_S = 2^4 - 1 = 15$):** $\eta = \frac{15}{63.5} = 0.2362 = \mathbf{23.62\%}$.
- **3. Selective Repeat ($k = 4 \implies W_S = 2^{4-1} = 8$):** $\eta = \frac{8}{63.5} = 0.1259 = \mathbf{12.60\%}$.
- **Optimal Window Size:** $W_{\text{optimal}} \geq 1 + 2a = 64 \implies$ Requires $k = 7$ bit sequence numbers for 100% efficiency!

M3 Active Recall & 10-Question University Exam Master Bank

Q1. Compare Stop-and-Wait, Go-Back-N, and Selective Repeat ARQ protocols across window sizes, buffer requirements, and efficiency. (10 Marks)

- **Stop-and-Wait:** $W_S = 1, W_R = 1$; 1-bit sequence; $\eta = 1/(1 + 2a)$; zero receiver buffer.
- **Go-Back-N:** $W_S = 2^k - 1, W_R = 1$; cumulative ACKs; retransmits entire window on loss; simple receiver.
- **Selective Repeat:** $W_S = 2^{k-1}, W_R = 2^{k-1}$; individual ACKs / NAKs; retransmits ONLY corrupted frame; receiver buffers out-of-order packets.

Q2. A bit stream '1101011011' is transmitted using standard CRC polynomial $x^4 + x + 1$. Generate the transmitted codeword. (10 Marks)

$G(x) = x^4 + x + 1 \implies 10011$ ($r = 4$). Append 4 zeros to data: '11010110110000'.

Performing Modulo-2 binary division by '10011' yields remainder $R = 1110$.

The transmitted codeword is **11010110111110**.

Q3. Derive the relationship between Minimum Hamming Distance d_{min} and the number of detectable and correctable errors. (8 Marks)

- **Detection of d errors:** No valid codeword can be transformed into another valid codeword by d bit flips $\implies d_{\text{min}} \geq d + 1$.
- **Correction of t errors:** Spheres of radius t centered at each codeword must be strictly non-overlapping $\implies d_{\text{min}} \geq 2t + 1$.

Q4. Explain Bit Stuffing in HDLC. What happens when the bitstream '0111110111110' is transmitted? (6 Marks)

In HDLC, sender inserts a '0' after five consecutive '1's. The input '0 11111 10 11111 0' becomes '0 11111 0 10 11111 0 0'. The receiver strips the stuffed zeros to recover the original payload.

Q5. Explain the Internet Checksum algorithm used in IP/TCP headers with a 16-bit binary addition example. (8 Marks)

The sender divides data into 16-bit words, sums them using 1's complement arithmetic (adding end-around carry back to sum), and inverts all bits to form the checksum. The receiver sums all 16-bit words including the checksum; if no bit errors occurred, the final sum is all 1's ('0xFFFF').

Topic 27.2: Comprehensive Sliding Window Mathematical Derivations & Buffer Protocols

Mathematical Proof: Maximum Window Size for Go-Back-N ($W_S = 2^k - 1$)

Suppose $k = 2$ bit sequence numbers (0, 1, 2, 3). If the sender were allowed a window size of $W_S = 2^k = 4$:

1. Sender transmits frames 0, 1, 2, 3. Receiver receives all 4 frames and emits ACK 0 (expecting frame 0 next).
2. If all ACKs are lost in transit: Sender times out and retransmits frame 0.
3. **Fatal Ambiguity:** Receiver cannot determine whether this incoming frame 0 is a *retransmission of the old frame 0* or a *brand-new frame 0 of the next sequence cycle*!
4. To guarantee complete non-overlapping receiver expectations:

$$W_S + W_R \leq 2^k \implies W_S + 1 \leq 2^k \implies W_S \leq 2^k - 1.$$

Mathematical Proof: Maximum Window Size for Selective Repeat ($W_S = W_R = 2^{k-1}$)

In Selective Repeat, the receiver accepts out-of-order frames within its window W_R .

To prevent window overlap between old and new cycles: $W_S + W_R \leq 2^k$

Setting symmetric windows $W_S = W_R = N$: $2N \leq 2^k \implies N \leq 2^{k-1}$

For $k = 3$ bits (0...7): Maximum window size for Selective Repeat is $W_S = 2^{3-1} = 4$.

Solved Numerical: Stop-and-Wait ARQ Efficiency with Frame Transmission Errors

A 1 Mbps link with $T_{\text{prop}} = 20$ ms transmits 1000 byte frames. If the frame error probability is $P_f = 0.1$ (10% packet loss), calculate the effective throughput.

Solution:

$$T_{\text{trans}} = \frac{8000 \text{ bits}}{10^6 \text{ bps}} = 8 \text{ ms} \implies a = \frac{20 \text{ ms}}{8 \text{ ms}} = 2.5$$

$$\text{Ideal Efficiency: } \eta_0 = \frac{1}{1 + 2a} = \frac{1}{1 + 2(2.5)} = \frac{1}{6} \approx 16.67\%$$

$$\text{Efficiency with Error Rate } P_f: \eta = \eta_0 \times (1 - P_f) = \frac{1 - 0.1}{6} = \frac{0.9}{6} = 0.15 = 15\%$$

$$\text{Effective Throughput} = \eta \times \text{Bandwidth} = 0.15 \times 1 \text{ Mbps} = 150 \text{ kbps}$$

Topic 27.3: Advanced Error Control & Sliding Window State Automata

In Data Link Layer protocol engineering, reliable sliding window protocols are formally modeled as Finite State Machines (FSM) maintaining transmit and receive state variables:

Complete Step-by-Step Go-Back-N vs. Selective Repeat Execution Trace under Frame Loss

Suppose 3-bit sequence numbers ($0 \dots 7$) are used with window size $W_S = 4$. Frame 2 is corrupted/lost during transmission from Sender A to Receiver B :

Event #	Go-Back-N (GBN) Protocol Behavior	Selective Repeat (SR) Protocol Behavior
1. Transmit	A sends frames 0, 1, 2, 3. Frame 2 is lost in transit.	A sends frames 0, 1, 2, 3. Frame 2 is lost in transit.
2. Receive 0, 1	B receives 0, emits ACK 1. B receives 1, emits ACK 2.	B receives 0, emits ACK 0. B receives 1, emits ACK 1.
3. Receive 3	B receives frame 3. Since frame 2 is missing, B discards frame 3 and re-emits cumulative ACK 2.	B receives frame 3. B buffers frame 3 in memory and emits NAK 2 (or ACK 3).
4. Timeout / NAK	A 's timer for frame 2 expires. A must Go-Back-N and retransmit all 2, 3 (even though 3 was already sent).	A receives NAK 2. A retransmits ONLY frame 2 .
5. Delivery	B receives 2, delivers 2 to network layer; then receives 3, delivers 3.	B receives retransmitted frame 2, delivers buffered frames $\{2, 3\}$ to network layer instantly!

Conclusion: Selective Repeat achieves far superior channel utilization on noisy and high-latency links by avoiding redundant retransmissions!

Step-by-Step Two-Dimensional (2D) Parity Check Calculation

Data block of four 7-bit ASCII characters: '1001100', '1100110', '1010101', '0111001'. Construct Even Parity rows and columns:

```

Data Row 1:  1  0  0  1  1  0  0 |  1 (Row Parity Bit 1)
Data Row 2:  1  1  0  0  1  1  0 |  0 (Row Parity Bit 2)
Data Row 3:  1  0  1  0  1  0  1 |  0 (Row Parity Bit 3)
Data Row 4:  0  1  1  1  0  0  1 |  0 (Row Parity Bit 4)
-----
Col Parity:  1  0  0  0  1  1  0 |  1 (Corner Parity)

```

Error Detection & Correction: A single-bit flip anywhere in the matrix causes exactly one row parity violation and one column parity violation, pinpointing the intersection coordinates for instant 1-bit error correction!

Topic 27.4: Master University Solved Examination Numerical Bank

 Solved Numerical 4: CRC Hardware Shift-Register Circuit Implementation

Design the Linear Feedback Shift Register (LFSR) hardware circuit for generating CRC-4 with polynomial $G(x) = x^4 + x + 1$.

- The circuit consists of 4 D-Flip-Flops (FF_3, FF_2, FF_1, FF_0) initialized to zero.
- **XOR** gates are placed at positions corresponding to non-zero coefficients of $G(x)$ (specifically before FF_0 and between FF_0 and FF_1).
- Data bits are shifted in MSB first. After k shifts of data and $r = 4$ shifts of zeros, the contents of the 4 flip-flops contain the exact 4-bit CRC remainder!

Solved Numerical 5: Go-Back-N Window Size Optimization under High Propagation Delay

A 100 Mbps fiber-optic link connects two data centers separated by $D = 2000 \text{ km}$ ($v = 2 \times 10^8 \text{ m/s}$). Frame size $L = 1500 \text{ bytes}$ (12,000 bits). What is the minimum number of sequence number bits (k) required for Go-Back-N ARQ to achieve 100% link utilization?

Solution:

$$T_{\text{prop}} = \frac{2000 \times 10^3 \text{ m}}{2 \times 10^8 \text{ m/s}} = 0.010 \text{ s} = 10 \text{ ms}$$

$$T_{\text{trans}} = \frac{12,000 \text{ bits}}{100 \times 10^6 \text{ bps}} = 0.00012 \text{ s} = 0.12 \text{ ms}$$

$$a = \frac{T_{\text{prop}}}{T_{\text{trans}}} = \frac{10 \text{ ms}}{0.12 \text{ ms}} = 83.33 \Rightarrow 1 + 2a = 1 + 2(83.33) = \mathbf{167.67}$$

For 100% utilization, sender window must satisfy: $W_s \geq 1 + 2a = 168$

In GBN, $W_S = 2^k - 1 \geq 168 \implies 2^k \geq 169 \implies \mathbf{k = 8 \text{ bits}}$ ($2^8 - 1 = 255 \geq 168$)

Topic 27.5: HDLC Protocol Architecture & Mathematical Error Analysis

High-Level Data Link Control (HDLC) is the foundational bit-oriented Data Link Layer standard (ISO 13239) governing synchronous point-to-point and multipoint communication lines.

HDLC Frame Type	Control Field Encoding	Operational Purpose
I-Frame (Information)	<code>`0   N(S) [3 bits]   P/F   N(R) [3 bits]`</code>	Carries user network payload data with piggybacked sequence numbers $N(S)$ and acknowledgments $N(R)$.
S-Frame (Supervisory)	<code>`1 0   Type [2 bits]   P/F   N(R) [3 bits]`</code>	Flow and error control when no payload is ready: <code>`00`</code> =Receive Ready (RR), <code>`01`</code> =Receive Not Ready (RNR), <code>`10`</code> =Reject (REJ GBN), <code>`11`</code> =Selective Reject (SREJ).
U-Frame (Unnumbered)	<code>`1 1   Modifier [5 bits]   P/F`</code>	Link connection establishment and teardown: <code>`SABM`</code> (Set Asynchronous Balanced Mode), <code>`DISC`</code> (Disconnect), <code>`UA`</code> (Unnumbered Ack).

Step-by-Step Solved Problem: Comprehensive CRC-16 Polynomial Division Trace

Compute the CRC checksum for 16-bit dataword $D = 0xC4D1 = 1100010011010001$ using standard CRC-16-CCITT polynomial $G(x) = x^{16} + x^{12} + x^5 + 1 \implies 10001000000100001$ (degree $r = 16$).

- Append $r = 16$ zeros to D : $D \cdot 2^{16} = 11000100110100010000000000000000$.
- Perform Modulo-2 XOR binary division by the 17-bit divisor G .
- The 16-bit remainder generated is appended to D to produce the exact 32-bit transmitted codeword.
- *Error Detection Capability of CRC-16*: Detects 100% of single-bit errors, 100% of double-bit errors, 100% of odd-numbered bit errors, and 100% of burst errors of length ≤ 16 bits!

Solved Numerical 6: Checksum Calculation for IPv4 / UDP Packets

Calculate the 16-bit Internet Checksum for the following two 16-bit numbers: `0x66A2` and `0x3C42`. Show all binary additions and 1's complement end-around carries.

Solution:

```
  0110 0110 1010 0010  (0x66A2)
+ 0011 1100 0100 0010  (0x3C42)
-----
  1010 0010 1110 0100  (Sum = 0xA2E4, Carry = 0)

Invert all bits (1's complement):
~ 1010 0010 1110 0100 = 0101 1101 0001 1011 = 0x5D1B (Checksum)
```

Receiver Check: `0x66A2 + 0x3C42 + 0x5D1B = 0xFFFF` (All 1s $\implies$ Data Valid!).

Q6. Explain the HDLC Protocol Frame Format and the meaning of Poll/Final (P/F) bit. (8 Marks)

An HDLC frame consists of: `Flag (01111110)` [8b] $\rightarrow$ `Address` [8b] $\rightarrow$ `Control` [8/16b] $\rightarrow$ `Information Payload` [Variable] $\rightarrow$ `FCS Checksum (CRC-16)` [16b] $\rightarrow$ `Flag (01111110)` [8b].

The **Poll/Final (P/F) bit** serves dual roles: when sent by a primary station, $P = 1$ (Poll) solicits an immediate response from secondary stations; when sent by a secondary station, $F = 1$ (Final) indicates the concluding frame of its transmission response.

Q7. Explain Piggybacking in bidirectional sliding window protocols and its performance benefits. (6 Marks)

Piggybacking is the technique of attaching acknowledgment information (ACK sequence numbers) into outgoing data frames traveling in the reverse direction, rather than transmitting separate standalone ACK packets. It eliminates dedicated ACK frame headers and packet transmission overhead, significantly improving bidirectional channel efficiency!

Topic 27.6: Linear Block Codes & Parity-Check Matrix Mathematical Foundations

In formal algebraic coding theory, a **Linear (n, k) Block Code** maps k -bit data vectors $\mathbf{d} = (d_1, \dots, d_k)$ into n -bit codewords $\mathbf{c} = (c_1, \dots, c_n)$ via a **Generator Matrix** G :

$$\mathbf{c} = \mathbf{d} \cdot \mathbf{G} \quad (\text{Modulo-2 Matrix Multiplication})$$

Where $G = [I_k \mid P]$ is a $k \times n$ systematic generator matrix (I_k is $k \times k$ identity matrix, P is $k \times (n - k)$ parity submatrix).

Parity Check Matrix H & Syndrome Decoding Vector S :

$$H = [P^T \mid I_{n-k}] \quad ((n-k) \times n \text{ Matrix})$$

$$S = r \cdot H^T \quad (\text{Where } r = c \oplus e \text{ is the received vector})$$

- If $S = 0$: Zero errors detected $\implies$ Accept r . - If $S \neq 0$: The syndrome S matches the i -th column of H , identifying that bit position i contains the bit error!

Step-by-Step Solved Problem: Complete Systematic (7, 4) Hamming Code Matrix Trace

$$\text{Let } G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}. \text{ Encode dataword } d = [1, 0, 1, 1]:$$

$$c = [1, 0, 1, 1] \cdot G = [1, 0, 1, 1, 0, 1, 0]$$

Suppose received vector is $r = [1, 0, 0, 1, 0, 1, 0]$ (Bit 3 corrupted). Then:

$$S = r \cdot H^T = [0, 1, 1]^T \implies \text{Matches Column 3 of } H \implies \text{Error is in Bit 3! Flip Bit 3 to 1 to correct!}$$

Topic 27.7: Master University Long-Answer Exam Solutions Bank

Q8. Derive the Channel Utilization Efficiency (η) for Go-Back-N ARQ and Selective Repeat ARQ under non-zero transmission error probability (P_f). (10 Marks)

Let P_f be the frame error probability, and $a = T_{\text{prop}}/T_{\text{trans}}$:

• **Stop-and-Wait ARQ with Errors:** Average number of transmissions per frame is $N_{\text{trans}} = \frac{1}{1-P_f}$.

$$\eta_{\text{SW}} = \frac{1 - P_f}{1 + 2a}$$

• **Go-Back-N ARQ with Errors:** If a frame is lost, the entire window of $N = 1 + 2a$ frames must be retransmitted. The average number of transmitted frames per successful delivery is $1 + \frac{NP_f}{1-P_f}$.

$$\eta_{\text{GBN}} = \frac{1 - P_f}{1 + 2aP_f}$$

• **Selective Repeat ARQ with Errors:** Since ONLY the corrupted frame is retransmitted individually:

$$\eta_{\text{SR}} = 1 - P_f \quad (\text{Independent of propagation parameter } a!).$$

Conclusion: On high-latency satellite and fiber channels ($a \gg 1$), Selective Repeat is the only viable ARQ protocol!

Q9. Explain the Point-to-Point Protocol (PPP) frame structure, Link Control Protocol (LCP), and Network Control Protocol (NCP). (8 Marks)

PPP (RFC 1661) is the standard data link protocol for broadband DSL, dial-up, and leased lines:

1. **Framing:** Byte-oriented frame with `Flag (0x7E)`, `Address (0xFF)`, `Control (0x03)`, `Protocol (2B)`, `Payload`, `FCS (16/32b)`. Uses byte stuffing (`0x7D`).
2. **Link Control Protocol (LCP):** Establishes, configures, authenticates (PAP / CHAP), tests, and terminates the data link connection.
3. **Network Control Protocol (NCP):** Establishes and configures network layer protocols over the link (e.g. IPCP assigns IP addresses to clients dynamically).

Q10. Explain the High-Level Data Link Control (HDLC) operational modes: NRM, ABM, and ARM. (8 Marks)

HDLC supports three distinct operational station configurations:

1. **Normal Response Mode (NRM):** Unbalanced configuration with 1 Primary station and ≥ 1 Secondary stations. Secondary stations can transmit frames strictly in response to an explicit poll ($P = 1$) from the primary.
2. **Asynchronous Balanced Mode (ABM):** Balanced point-to-point configuration with 2 Combined stations. Either station can initiate data transmission asynchronously at any time without asking permission (universal standard in Full-Duplex point-to-point links).
3. **Asynchronous Response Mode (ARM):** Unbalanced configuration where Secondary stations can initiate transmission asynchronously without waiting for a primary poll.